

Time Series Forecasting for Portfolio Management

Final Investment Memo | GMF Investments | 2026-07-07

TSLA • BND • SPY • ARIMA • LSTM • Efficient Frontier • Backtesting

4.60%	LSTM test MAPE	-32.5%	TSLA 12mo forecast	0.85	Max Sharpe portfolio	1.69 VS 1.66	Strategy vs. benchmark Sharpe
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Executive Summary

This memo presents GMF Investments' time series forecasting and portfolio optimization analysis for a three-asset portfolio: **TSLA** (high-growth equity), **BND** (investment-grade bonds), and **SPY** (broad market index), using daily data from 2015-01-01 to 2026-06-30. We cleaned and explored the data, built and compared ARIMA and LSTM forecasting models for TSLA, generated a 12-month forward forecast with confidence intervals, used Modern Portfolio Theory to derive an optimal three-asset portfolio, and backtested that portfolio against a passive 60/40 SPY/BND benchmark. Consistent with the Efficient Market Hypothesis, we find limited exploitable signal in TSLA's historical prices alone; our recommendation therefore treats the forecast as one input among several, not a standalone trading signal.

Task 1 — Data Preprocessing and Exploratory Analysis

Daily OHLCV data for TSLA, BND, and SPY were fetched via yfinance for 2015-01-01 through 2026-06-30, returning 2,888 trading days per ticker with no missing values. Prices were reindexed onto a common business-day calendar and any gaps were forward-filled (not interpolated), since a price is genuinely unchanged until the next trade occurs.



Figure 1. Adjusted close price, actual and normalized.

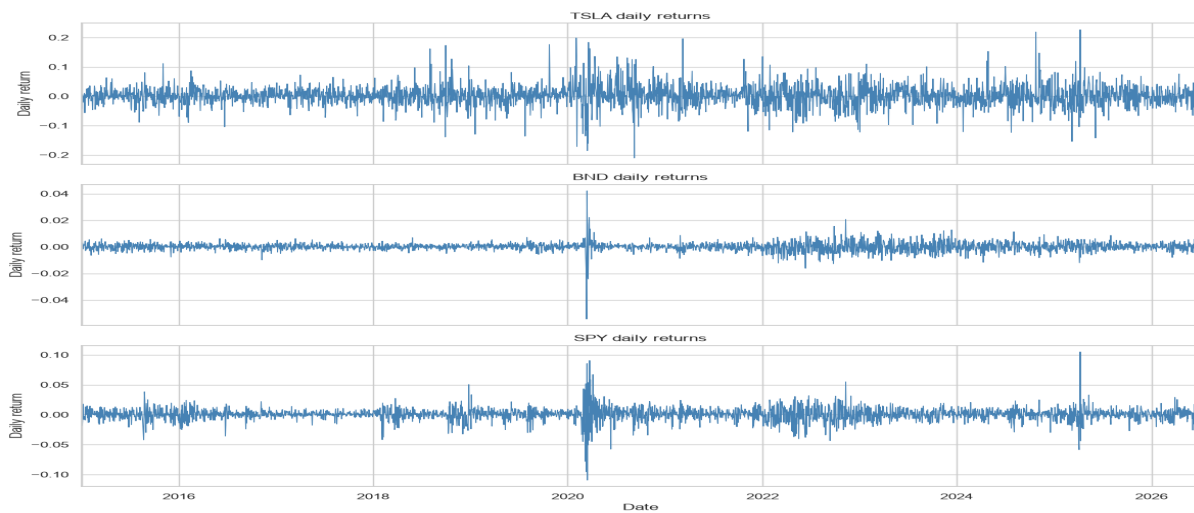


Figure 2. Daily percentage returns by asset.

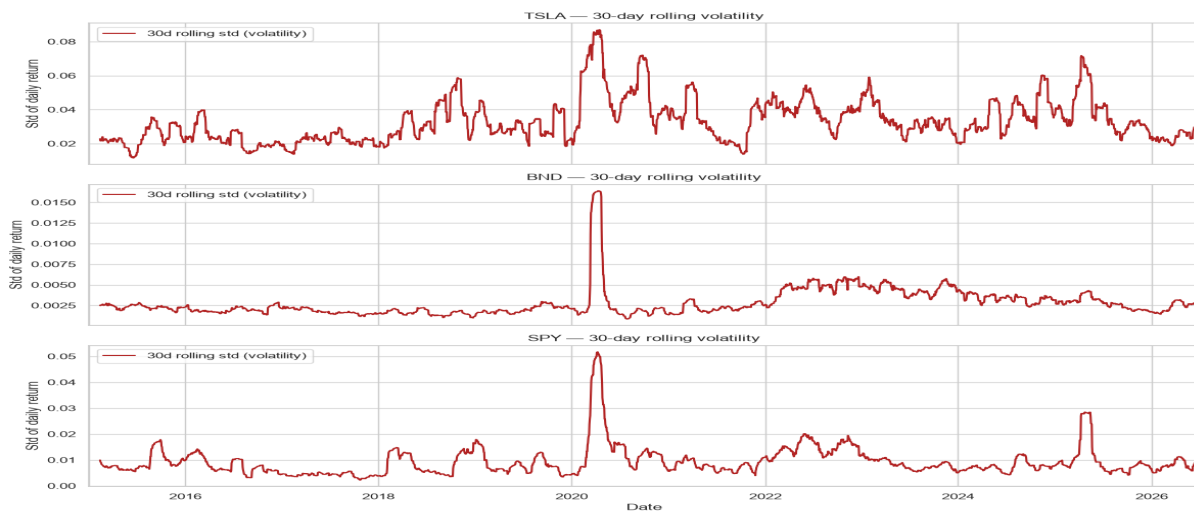


Figure 3. 30-day rolling volatility.

Stationarity testing (Augmented Dickey-Fuller)

Series	ADF statistic	p-value	Stationary at 5%?
TSLA — Close (level)	-1.04	0.739	No
TSLA — Daily return	-55.15	<0.001	Yes
BND — Close (level)	-1.12	0.708	No
BND — Daily return	-21.28	<0.001	Yes
SPY — Close (level)	1.43	0.997	No
SPY — Daily return	-15.70	<0.001	Yes

All three price *levels* fail to reject the ADF null (non-stationary, random-walk-like), consistent with the Efficient Market Hypothesis; all daily *return* series are stationary, confirming a first difference ($d=1$) is required for ARIMA.

Risk metrics (annualized, 0% risk-free rate)

Asset	Ann. Return	Ann. Volatility	VaR (95%, daily)	Sharpe
TSLA	43.8%	56.1%	5.11%	0.78
BND	1.9%	5.2%	0.47%	0.37
SPY	13.9%	17.3%	1.64%	0.80

Task 2 — Forecasting Models: ARIMA vs. LSTM

The TSLA adjusted-close series was split chronologically (no shuffling): train = 2015-01-02 to 2024-12-31 (2,608 observations), test = 2025-01-01 to 2026-06-29 (391 observations). ARIMA: `pmdarima.auto_arma` selected ARIMA(0,1,0) — a driftless random walk — forecast statically over the full test horizon. LSTM: 60-day lookback window, two stacked LSTM(50) layers with dropout, trained with early stopping; test predictions use the true preceding 60 days at each step (one-step-ahead evaluation).

Model	MAE	RMSE	MAPE (%)	Forecast mode
ARIMA(0,1,0)	54.15	70.20	17.11	Static, unconditional, 391-day
LSTM (60-day window)	17.22	21.34	4.60	One-step-ahead, true history

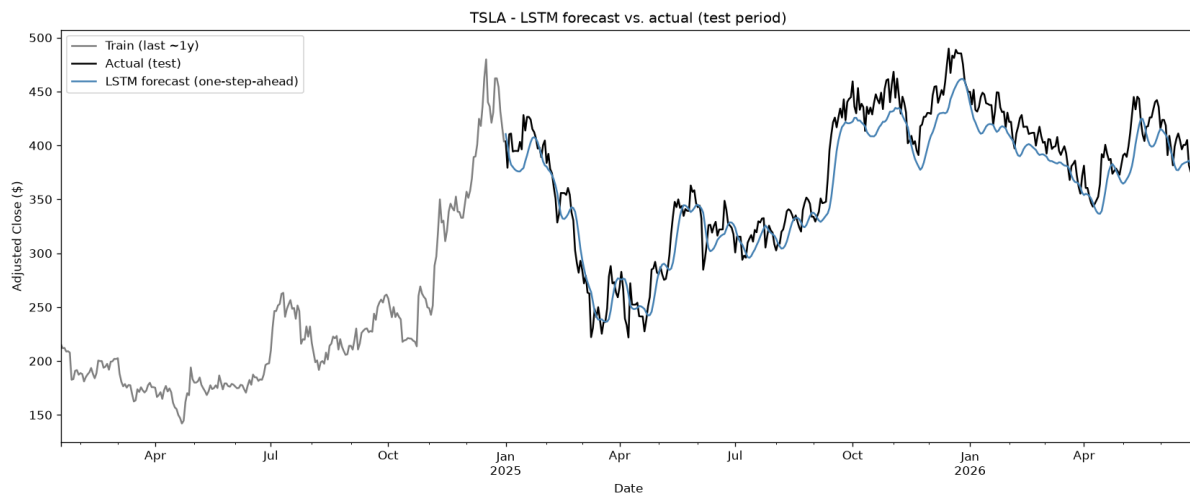


Figure 4. LSTM one-step-ahead forecast vs. actual, test period.

These are not a perfectly apples-to-apples comparison — ARIMA forecasts the full 18-month horizon unconditionally, while the LSTM is re-conditioned on true history at every step (an inherently easier task). Both models agree TSLA price levels are close to a random walk with little linear autocorrelation structure, consistent with the Efficient Market Hypothesis. We select the **LSTM** to carry forward into Task 3.

Task 3 — Forecasting Future Market Trends

A genuine 12-month (252 trading day) **recursive** LSTM forecast was generated (each day's prediction feeds back in as the next day's context), with a 95% confidence band from Monte Carlo simulation calibrated to TSLA's historical drift/volatility, multiplicatively re-centered on the LSTM's own trajectory so the band can never cross zero.

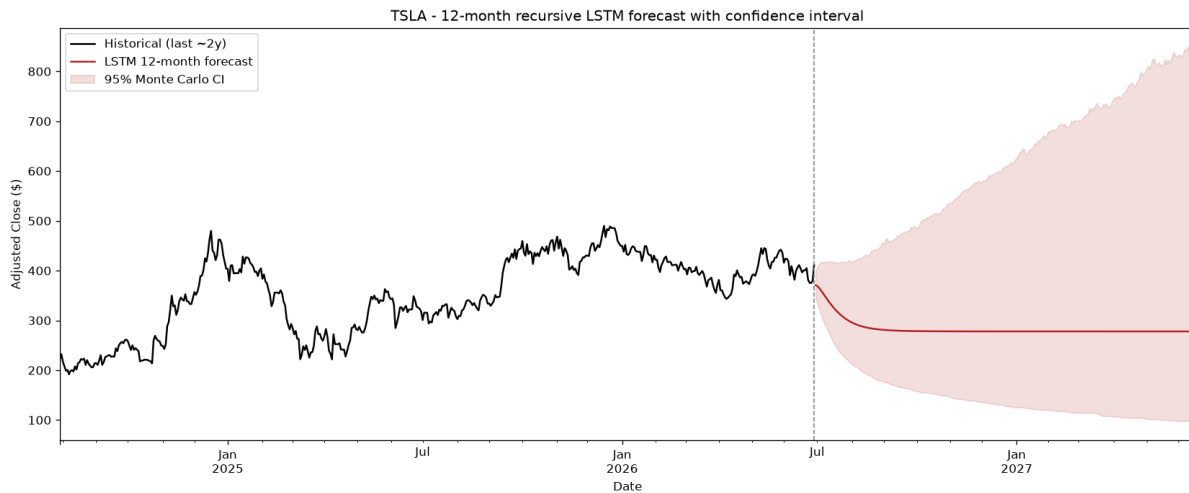


Figure 5. 12-month recursive LSTM forecast with 95% Monte Carlo CI.

Last observed price (2026-06-29): \$411.84 12-month point forecast: \$277.90 (implied return: -32.5%)
95% CI at 12 months: [\$97.57, \$846.63]

A fixed-point artifact, not a bearish thesis. The forecast drops from \$411.84 toward the high-\$270s over the first ~2 months, then converges to a fixed point (~\$277.90) and stays essentially flat for the remaining ~10 months. This is a well-documented failure mode of naive recursive one-step LSTM forecasting: once consecutive model-generated predictions stop changing, the recursion has settled into an attractor and simply reproduces it. We treat the -32.5% point estimate with real skepticism rather than at face value — it reflects the early collapse, not a considered 12-month view. The 95% interval itself widens from ~\$57 at day 1 to ~\$749 by month 12, an independent signal that long-horizon certainty is low.

Task 4 — Portfolio Optimization (Modern Portfolio Theory)

TSLA's expected return uses the Task 3 LSTM forecast (-32.5% annualized); BND and SPY use historical mean annualized daily returns (1.9% and 13.9%). The covariance matrix uses sample covariance of historical daily returns for all three assets, annualized.

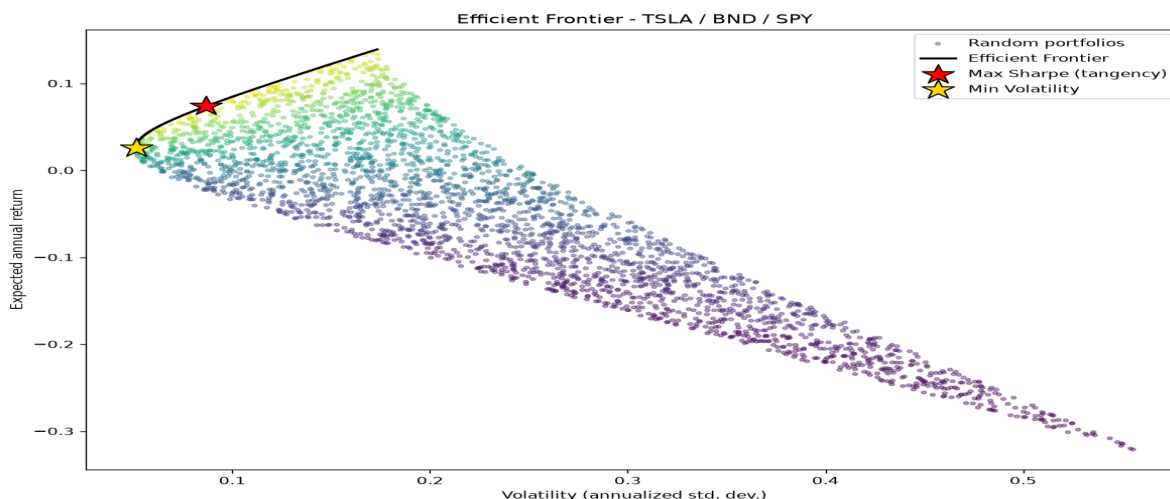


Figure 6. Efficient Frontier — TSLA / BND / SPY.

Portfolio	TSLA	BND	SPY	Exp. Return	Volatility	Sharpe
Max Sharpe (recommended)	0.0%	54.7%	45.3%	7.35%	8.66%	0.85
Min Volatility	0.0%	94.5%	5.5%	2.58%	5.13%	0.50

Both optimal portfolios allocate **0% to TSLA**: given its negative forecasted return combined with by-far-highest volatility (~56% annualized) and only moderate diversification benefit (~0.49 correlation with SPY), no efficient portfolio wants TSLA exposure at this input. We recommend the **Maximum Sharpe Ratio portfolio** (54.7% BND / 45.3% SPY) as the standard institutional default absent a client-specific risk mandate. This recommendation is explicitly conditional on the Task 3 TSLA view — a more optimistic forecast would likely restore some TSLA allocation, illustrating how directly forecast quality drives the downstream portfolio decision.

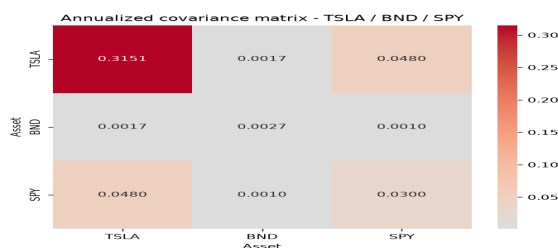


Figure 7. Annualized covariance matrix.

Task 5 — Strategy Backtesting

The Max Sharpe portfolio (monthly rebalanced) was simulated over the last 12 months of the dataset (2025-07-01 to 2026-06-29) against a static 60% SPY / 40% BND benchmark.

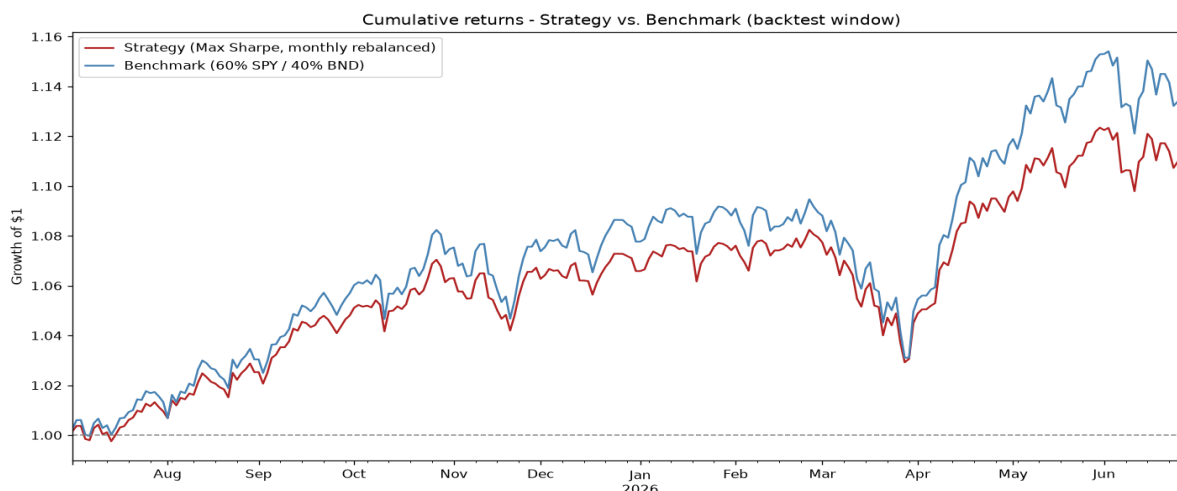


Figure 8. Cumulative returns — strategy vs. benchmark.

Metric	Strategy (Max Sharpe)	Benchmark (60/40)
Total Return	11.64%	14.21%
Annualized Return	11.31%	13.80%
Annualized Volatility	6.48%	7.96%
Sharpe Ratio	1.69	1.66
Max Drawdown	-4.91%	-5.82%

No unambiguous win either way. The benchmark delivered higher raw return (more SPY exposure in a rising market); the strategy edged out the benchmark on risk-adjusted terms (higher Sharpe, smaller drawdown) by holding more BND and zero TSLA. In short: the strategy was the steadier ride, the benchmark was the better raw return, in this particular year — a direct consequence of Task 3's negative TSLA view driving Task 4's allocation. A single 12-month backtest with this narrow a margin is far too little evidence to call this proof of durable outperformance in either direction.

Overall Conclusion

This project demonstrates the full pipeline from raw price data to a backtested, model-informed portfolio recommendation. The central, recurring finding across every task is consistent with the Efficient Market Hypothesis: TSLA's price history contains limited robust, exploitable structure, and both classical (ARIMA) and deep learning (LSTM) models largely rediscover this fact rather than refuting it — the LSTM's own recursive forecast collapsing to a fixed point is a vivid illustration. Forecast uncertainty compounds quickly beyond a few weeks, and any portfolio decision built on these forecasts should treat them as one input among several — combined with historical risk data on BND/SPY and sound portfolio construction (MPT) — rather than as precise, standalone predictions. GMF's recommended allocation (54.7% BND / 45.3% SPY, 0% TSLA) reflects a risk-adjusted balance validated, with meaningful caveats, by a one-year backtest against a passive benchmark.